

The PiLLAR System: Positional Addressing in the Digits of Pi

TZPID Companion Spine Series, Paper C13 of XXVII
Logarithmic address retrieval, PiID, and the TZPi index

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Abstract

This companion introduces the Pi Linear Logarithmic Address Retrieval (PiLLAR) system: a positional identification architecture over the decimal expansion of pi, with a logarithmic bound for unique local address retrieval, a 13-digit positional identifier (PiID), and auxiliary structures including the TZPi index. The source manuscript’s stated thesis: “This paper introduces the Pi Linear Logarithmic Address Retrieval System (PiLLAR System), a positional identification architecture defined over the decimal expansion of pi. The system establishes a logarithmic bound for unique local address retrieval within finite segments of pi, defines a 13-digit positional identifier termed PiID, and introduces auxiliary structures including TZPi (Trawin Zero Point index), PiPE-Line . . .” The paper organizes this thesis as a TZPID gold spine: a curated seven-node registry chain (ID8559, ID0428, ID9010, ID6190, ID3532, ID9009, ID8896) with typed proof-obligation roles, dictionary and encyclopedia anchors for every node, a declared dependency order, and stubs in the program’s Isabelle/Wolfram verification pipeline. Within the arc this paper supplies the program’s addressing layer: how structures inside an unbounded irrational expansion are located, named, and retrieved.

1 Introduction

The TZPID program is, at bottom, a registry: ten thousand identified objects requiring stable addresses. The PiLLAR system is the corpus’s most radical proposal for what an address can be: a position in the decimal expansion of π . The system establishes a logarithmic bound for unique local address retrieval within finite segments of the expansion—the depth one must read before a local digit window becomes unique—and packages the result as a 13-digit positional identifier, the PiID, with auxiliary structures (the TZPi index, the PiPE-Line retrieval protocol) completing the architecture.

The mathematical footing is the expected normality of π : in a normal expansion every finite digit block occurs with its combinatorial frequency, so a window of length $\sim \log_{10} N$ suffices to address position N uniquely within a segment—the logarithmic bound that gives the system its name. The 13-digit choice calibrates the bound to registry scale with comfortable margin.

Why house this in a physics corpus? Because the program’s deepest numerical structures are themselves π -laden: the enclosure’s eigenvalue ladder is anchored at $n\pi$ (core Paper I), the allowed orbital ratios populate a π -ladder (ID7904), and the unified charge of Paper VIII carries π explicitly in its winding term. A registry that addresses its objects by positions in π is practicing what its physics preaches: that the constant is not decoration but infrastructure. The spine below anchors the addressing chain; refinement targets include collision statistics for PiID at registry scale and formalization of the retrieval bound.

2 The concept-anchored spine methodology

The companion series applies a uniform method, stated here so that the paper is self-contained. The source manuscript was published with its mathematics rendered as text rather than as machine-readable LaTeX, so its spine is *concept-anchored*: the thesis is taken from the manuscript’s abstract, and the spine nodes are the canonical-registry entries most strongly matched to the manuscript’s themes, selected by weighted term overlap with foundational nodes ranked first. Each node carries its registry equation, its dictionary definition, its encyclopedia interpretation, and a declared proof-obligation role (Core_Definition, Core_Axiom, or Derived_Theorem_Obligation). The resulting chain is a starting backbone for refinement—a typed scaffold onto which an exact, equation-by-equation crosswalk with the source manuscript can be progressively attached—rather than a claim of completed formalization.

Three properties make the backbone useful even before refinement completes. First, *addressability*: every claim in the companion paper resolves to a registry identifier whose equation, definition, and verification status are independently inspectable, so disagreement can be localized to a node rather than to the paper as a whole. Second, *pipeline coupling*: each spine is paired with an Isabelle focus-theory stub and a Wolfram check stub, so that as nodes are refined their obligations enter the same machine-checked pipeline that certifies the core series’ guardrails. Third, *role discipline*: the obligation roles separate what the spine defines from what it asserts and what it must prove, which keeps the demarcation section of every companion paper honest by construction.

The dependency chain should accordingly be read as a proof-order proposal: upstream nodes supply the vocabulary and axioms that downstream nodes consume, and the refinement pass is free to reorder where the source manuscript’s own logic demands it. Where a node’s registry equation arrives with rendering artifacts inherited from the source extraction (a known cost of text-rendered mathematics), the equation is quoted as carried by the registry, with the dictionary and encyclopedia entries supplying the authoritative reading.

3 The registry chain

The curated chain, in proof order, is

$$\begin{aligned} \text{ID8559} &\rightarrow \text{ID0428} \rightarrow \text{ID9010} \rightarrow \text{ID6190} \\ &\rightarrow \text{ID3532} \rightarrow \text{ID9009} \rightarrow \text{ID8896}. \end{aligned}$$

We take the nodes in order; each subsection quotes the registry equation as carried by the canonical master, then gives the dictionary definition and the encyclopedia’s interpretive reading.

3.1 ID8559: Definition of V (Core_Definition)

Registry equation:

$$v=1(Vv(x)S9600)$$

Dictionary definition. Definition of V — $v=1(Vv(x)S9600)$

Encyclopedia reading. Definition of V is an algebraic definition, written $v=1(Vv(x)S9600)$. It is an algebraic definition expressing the quantity directly in terms of the variables on its right-hand side. It is built from V, S, defining v. Within the Trawin Zero Point registry it serves as one element of the framework’s mathematical narrative.

Computational guardrail: pi_slide_ID8559_identity.

Obligation reading. As a Core_Definition this node contributes vocabulary rather than assertion: downstream nodes may use its object freely, but nothing is yet claimed about the world. Its refinement burden is precision—the typed form must capture exactly the object the source manuscript deploys.

3.2 ID0428: Toroidal Flux-Gating & Channel Segregation (Core_Definition)

Registry equation:

$$\mathcal{I}(T:A:M) = \cdots$$

Dictionary definition. ID 428 establishes the tripartite information architecture that emerges when energy is extracted at a Trawin Zero Point.

Encyclopedia reading. T: Canonical Definition is localized within the manifold and metric dynamics arc of the directory. R: Aggregate Relation preserves the operative relation that berry phase tomography & atemporal readout requires. A: Closure Relation fixes the admissible closure condition for the entry. W: BERRY PHASE TOMOGRAPHY & ATEMPORAL READOUT is rendered as a reusable operator-level state- ment. I: The informational content of berry phase tomography & atemporal readout is retained rather than flattened. N: BERRY PHASE TOMOGRAPHY & ATEMPORAL READOUT closes as a distinct normalized TRAWIN ob- ject. Interpretive role: This entry specifies the structure within the registry.

Computational guardrail: pi_slide_ID0428_identity.

Obligation reading. This `Core_Definition` fixes an object the rest of the chain consumes. In proof order it must be discharged first; in refinement it is where a mismatch between the manuscript’s usage and the registry’s typing would first surface.

3.3 ID9010: Relation: a I alpha > epsilon (Core_Axiom)

Registry equation:

$$\mathcal{A}_{\alpha} > \epsilon$$

Dictionary definition. Relation: $a \models \alpha$ — $A \models \alpha$

Encyclopedia reading. Relation: $a \propto \epsilon$ is an inequality / bound, written $A \propto \epsilon$. It states an inequality or bound that constrains the admissible range of the quantity. It is built from α , ϵ , A . Within the Trawin Zero Point registry it serves as one element of the framework's mathematical narrative.

Computational guardrail: pi_slide_ID9010_threshold.

Obligation reading. As a Core_Axiom this node is where the spine commits: it asserts without deriving, and everything downstream inherits the commitment. The demarcation section flags it accordingly, and the refinement pass should seek either an upstream derivation or an explicit empirical anchor.

3.4 ID6190: Full Two-Particle Ansatz (With Toroidal Relative Shape) (Derived_Theorem_Obligation)

Registry equation:

$$\boxed{\Psi(\mathbf{r}, t) = \int_{-\infty}^{\infty} \tilde{A}(\mathbf{r}, \omega) e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})} d\omega}$$

Dictionary definition. Full Two-Particle Ansatz (With Toroidal Relative Shape) — $\Psi(\mathbf{R}, \mathbf{r}, t) = \mathcal{N} F(\mathbf{R}) G(\mathbf{r}, t; \sigma) \exp\left[i \int_{-\infty}^t dt' \int d\mathbf{p} \, \mathbf{p} \cdot \mathbf{r} \, A(\mathbf{p}) \right]$

Encyclopedia reading. Full Two-Particle Ansatz (With Toroidal Relative Shape) is a commutator / bracket relation, written $\Psi(\mathbf{R}, \mathbf{r}, t) = \int_{-\infty}^{\infty} d\mathbf{r}_0 \exp\left(\frac{i}{\hbar} \int_{t_0}^t dt' \mathcal{H}(\mathbf{R}, \mathbf{r}, t')\right) \Psi(\mathbf{R}, \mathbf{r}_0, t_0)$. It is built from $\Psi, \ell, \theta, \mathbf{R}, \mathbf{r}$. Within the Trawin Zero Point registry it serves as one element of the framework's mathematical narrative.

Computational guardrail: pi_slide_ID6190_identity.

Obligation reading. As a Derived_Theorem_Obligation this node owes a proof: the registry asserts that it follows from the chain's upstream vocabulary and axioms, and the Isabelle focus stub holds the slot where that derivation will be discharged.

3.5 ID3532: Quantum Quantum State Functional (Derived_Theorem_Obligation)

Registry equation:

$$\left[\boxed{\Psi(\mathbf{r}, t) = \mathcal{N} \int_{-\infty}^{\infty} G_l(\mathbf{r}, \theta; \sigma_r(t)) e^{i \frac{p(\mathbf{r} + \mathbf{x}_0)}{\hbar}} \tilde{A}(p) dp} \right]$$

Dictionary definition. Quantum Quantum State Functional — $\Psi(\mathbf{r}, \mathbf{r}, t) = \mathcal{N} \int_{-\infty}^{\infty} G_l(\mathbf{r}, \theta; \sigma_r(t)) e^{i \frac{p(\mathbf{r} + \mathbf{x}_0)}{\hbar}} \tilde{A}(p) dp$

Encyclopedia reading. The entry preserves the equation exactly as extracted from 'D:\00_EngineCSVconcept_units.csv' for registry alignment and later derivation.

Computational guardrail: pi_slide_ID3532_identity.

Obligation reading. A derived obligation: the spine claims this follows from what precedes it. Until the formal derivation is supplied, the computational guardrail polices at least the node's internal consistency.

3.6 ID9009: Definition of Phi(p) (Derived_Theorem_Obligation)

Registry equation:

$$\Phi(\mathbf{p}) = \sum_i A_i \Phi_i$$

Dictionary definition. Definition of Phi(p) — $\Phi(p) = \sum_i A_i \Phi_i$

Encyclopedia reading. Definition of Phi(p) is a series / summation, written $\Phi(p) = \sum_i A_i \Phi_i$. It is expressed as a sum or series over modes or indices, superposing discrete contributions. It is built from alpha, Phi, A, defining Phi(p). Within the Trawin Zero Point registry it serves as one element of the framework's mathematical narrative.

Computational guardrail: pi_slide_ID9009_identity.

Obligation reading. This node carries proof debt by declaration—Derived_Theorem_Obligation—and is therefore where the refinement pass should concentrate first: a discharged derivation here strengthens every downstream node simultaneously.

3.7 ID8896: DECODING/LOOKUP TIME COMPLEXITY Canonical Statutory Formula: T_dec (Derived_Theorem_Obligation)

Registry equation:

$$T_{\text{dec}} = O(\log 153600) \approx O(1)$$

Dictionary definition. DECODING/LOOKUP TIME COMPLEXITY Canonical Statutory Formula: $T_{\text{dec}} = O(\log 153600) \approx O(1)$ Formal Technical Dissertation: Formalizes the retrieval speed of the system.

Encyclopedia reading. DECODING/LOOKUP TIME COMPLEXITY Canonical Statutory Formula: T_{dec} is a scaling / order-of-magnitude relation, written $T_{\text{dec}} = O(\log 153600) \approx O(1)$. It expresses a scaling or proportionality, capturing how the quantity grows with its parameters. It is built from O, defining T_{dec} . Within the Trawin Zero Point registry it serves as one element of the framework's mathematical narrative.

Computational guardrail: `pi_slide_ID8896_identity`.

Obligation reading. As a `Derived_Theorem_Obligation` this node owes a proof: the registry asserts that it follows from the chain’s upstream vocabulary and axioms, and the Isabelle focus stub holds the slot where that derivation will be discharged.

4 Dependency structure and obligations

Table 1 summarizes the spine’s obligation ledger. The chain’s proof order runs from the definitional nodes (vocabulary first) through the derived obligations to the spine’s closure; the refinement pass may interleave additional registry nodes where the source manuscript’s own derivations require them, in which case the table grows monotonically—existing rows are never weakened, only joined.

Table 1: Obligation ledger for the pi slide spine.

ID	Obligation role	Wolfram check
ID8559	Core_Definition	<code>pi_slide_ID8559_identity</code>
ID0428	Core_Definition	<code>pi_slide_ID0428_identity</code>
ID9010	Core_Axiom	<code>pi_slide_ID9010_threshold</code>
ID6190	Derived_Theorem_Obligation	<code>pi_slide_ID6190_identity</code>
ID3532	Derived_Theorem_Obligation	<code>pi_slide_ID3532_identity</code>
ID9009	Derived_Theorem_Obligation	<code>pi_slide_ID9009_identity</code>
ID8896	Derived_Theorem_Obligation	<code>pi_slide_ID8896_identity</code>

5 Crosswalk to the core series and companion corpus

Because all spines draw on one canonical registry, node sharing is measurable and meaningful: a shared identifier means two papers consume literally the same typed object, so verification work transfers between them without translation. Of this spine’s seven nodes, 0 appear in core-series spines and 0 recur elsewhere in the companion corpus. Table 2 records the census.

Table 2: Node-sharing census for this spine.

ID	Core-series appearances	Companion-corpus appearances
ID8559	—	—
ID0428	—	—
ID9010	—	—
ID6190	—	—
ID3532	—	—
ID9009	—	—
ID8896	—	—

Two readings follow. Nodes shared with the core series are this spine’s strongest members: their guardrails are already certified in the core pipeline, and the present paper inherits that status. Nodes unique to this spine are its distinctive contribution—and its refinement priority, since no other paper will discharge them. Nodes recurring across companions mark the corpus’s internal seams: the same object consumed by several manuscripts, whose refinements must agree by the duplicate discipline documented in the Einstein Focus companion.

6 Position in the TZPID arc

PiLLAR is infrastructure for the whole registry, but its conceptual kinship is with the ladder structures of core Papers I–II: the same constant that quantizes the enclosure’s spectrum here quantizes the address space. The TZPi index ties addressing to the program’s anchor point, and the logarithmic retrieval bound is the addressing-layer cousin of the program’s recurring economy claim—that structure is located by short codes because the underlying space is uniform.

7 Verification pathway

The spine enters the program’s two-engine verification pipeline. The Isabelle focus theory `TZPID_Spine_pi_slide.thy` types the seven obligations over the registry’s declared vocabulary, in the dependency order of Section 3; the Wolfram check stub `pi_slide_checks.wl` carries the per-node computational guardrails listed in Table 1. As with the core series, the division of labor is deliberate: the theorem prover certifies structure (types, roles, dependency soundness), while the computational engine certifies arithmetic and algebraic identities node by node. A check name of the form `..._identity` denotes a per-node identity guardrail generated from the registry equation; refined checks replace these as the equation-level crosswalk matures.

Because the companion spines share the registry with the core series, verification is cumulative: any node here that also appears in a core spine (the chain tables record several) inherits the core guardrail’s status, and any refinement proved here propagates back. The companion series thus functions as the proof pipeline’s breadth dimension—161 distinct registry nodes across the 27 companions—complementing the core series’ depth.

The pipeline’s two-engine division also fixes what failure means at this layer. A failed Wolfram guardrail localizes an arithmetic or algebraic defect in one node’s registered equation: the repair is editorial (correct the registration against the source) or substantive (the source itself errs), and the obligations ledger records which. A failed Isabelle discharge, by contrast, indicts the chain: a type mismatch or unprovable dependency means the proof-order proposal of Section `refsec:chain` is wrong as stated, and the refinement pass must re-curate. Keeping these failure modes separate is what the stub architecture buys: every companion enters the pipeline with its failure semantics declared in advance, which is the practical meaning of being peer-reviewable by machine.

8 Refinement worksheet

The concept-anchored backbone converts into an equation-level formalization through a bounded, node-by-node worksheet. For each node the worksheet item below states the concrete refinement act; completing all seven closes the spine’s first formalization pass.

1. **ID8559** (`pi_slide_ID8559_identity`): exhibit the source manuscript’s working definition beside the registry equation; reconcile notation; type the object in the focus theory and confirm every downstream node consumes the typed form.
2. **ID0428** (`pi_slide_ID0428_identity`): exhibit the source manuscript’s working definition beside the registry equation; reconcile notation; type the object in the focus theory and confirm every downstream node consumes the typed form.
3. **ID9010** (`pi_slide_ID9010_threshold`): state the axiom with its quantifiers explicit; attach the empirical anchor or upstream derivation candidate; declare the tolerance under which data could falsify it.

4. **ID6190** (`pi_slide_ID6190_identity`): reconstruct the source manuscript’s derivation sketch; identify the upstream nodes it actually uses (amending the chain if they differ from the curated order); discharge the proof in the focus theory or record the precise gap.
5. **ID3532** (`pi_slide_ID3532_identity`): reconstruct the source manuscript’s derivation sketch; identify the upstream nodes it actually uses (amending the chain if they differ from the curated order); discharge the proof in the focus theory or record the precise gap.
6. **ID9009** (`pi_slide_ID9009_identity`): reconstruct the source manuscript’s derivation sketch; identify the upstream nodes it actually uses (amending the chain if they differ from the curated order); discharge the proof in the focus theory or record the precise gap.
7. **ID8896** (`pi_slide_ID8896_identity`): reconstruct the source manuscript’s derivation sketch; identify the upstream nodes it actually uses (amending the chain if they differ from the curated order); discharge the proof in the focus theory or record the precise gap.

The worksheet’s order matters: definitional items unblock derived ones, and a failure at any item halts propagation rather than silently weakening downstream claims. Completed items should update the obligations CSV in place, so that the spine’s public ledger always reflects its true refinement state—the same live-ledger discipline the core series applies to its guardrails.

9 Demarcation and refinement targets

As throughout the program, the demarcation line is drawn explicitly. The established layer comprises the standard mathematics and physics that the chain’s nodes quote—definitions, identities, and independently citable results—together with whatever the computational guardrails certify. The framework layer comprises the source manuscript’s interpretive claims and the spine’s structural assertion that its nodes form the stated dependency chain. The retrieval bound is conditional on normality, which remains unproven for pi; the system’s practical claims (collision rates, retrieval depth) are empirical over computed expansions and are testable to any desired scale by direct computation.

Refinement targets, in pipeline order: (i) replace the concept-anchored node selection with an equation-level crosswalk against the source manuscript; (ii) promote each `..._identity` guardrail to a content-bearing check; (iii) discharge the typed obligations in the Isabelle focus theory against the program’s shared vocabulary; and (iv) record any node whose refinement fails, since a failed node localizes exactly where the source manuscript and the canonical registry disagree—the information a refinement pass exists to produce.

10 Conclusion

The PiLLAR system treats pi as the registry’s native soil: addresses as positions, retrieval as digit-window matching, uniqueness priced logarithmically. Its refinement is computational—collision audits at scale, bound formalization under explicit normality assumptions—and its reward is an addressing layer with the same flavor as the physics it serves: discrete identifiers living in a continuum structured by one constant.

References

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